

Object-Oriented Analysis and Design

Code Labs 加分題

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```
if (MoveCode == 1 && Characters[clientno - 1].y - 1 >= 0)
{ //up and walk
    Characters[clientno - 1].y = Characters[clientno - 1].y - 1;
    System.out.println("第"+clientno+"位玩家方向向上走");
    System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
    System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
    Characters[clientno - 1].dir = 1;
}
else if (MoveCode == 0 && Characters[clientno - 1].y + 1 <= 19)
{ //down and walk
    Characters[clientno - 1].y = Characters[clientno - 1].y + 1;
    System.out.println("第"+clientno+"位玩家方向向下走");
    System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
    System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
    Characters[clientno - 1].dir = 0;
}
else if (MoveCode == 2 && Characters[clientno - 1].x - 1 >= 0)
{ //left and walk
    Characters[clientno - 1].x = Characters[clientno - 1].x - 1;
    System.out.println("第"+clientno+"位玩家方向向左走");
    System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
    System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
    Characters[clientno - 1].dir = 2;
}
else if (MoveCode == 3 && Characters[clientno - 1].x + 1 <= 19)
{ //right and walk
    Characters[clientno - 1].x = Characters[clientno - 1].x + 1;
    System.out.println("第"+clientno+"位玩家方向向右走");
    System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
    System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
    Characters[clientno - 1].dir = 3;
}
```

以上程式可以用下方程式碼替代：

// 1. 定義移動對照表 {dx, dy, dir}，索引對應 MoveCode 0=下, 1=上, 2=左, 3=右

```
int[][] moveTable = {{0,1,0}, {0,-1,1}, {-1,0,2}, {1,0,3}};
```

```
int[] p = moveTable[MoveCode];
```

// 2. 預判新位置並在符合邊界 (0-19) 時更新座標與面向

```
int nx = Characters[clientno-1].x + p[0], ny = Characters[clientno-1].y + p[1];
```

```
if (nx >= 0 && nx <= 19 && ny >= 0 && ny <= 19) { Characters[clientno-1].x = nx;
```

```
Characters[clientno-1].y = ny; Characters[clientno-1].dir = p[2]; }
```